

KOBELYAKY, Ukraine

WMO: 336210

Lat: 49.1539N

Lon: 34.2078E

Elev: 387

StdP: 14.49

Time Zone: 2.00 (EUE)

Period: 84-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	2.6	-7.7	3.7	-1.6	-3.0	4.7	2.9	18.6	15.9	16.6	26.1	5.0	180	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	89.0	68.3	85.9	66.6	82.9	65.5	70.8	83.2	69.0	81.1	67.4	78.9	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	99.8	76.8	64.7	92.9	74.5	63.1	87.6	72.9	34.9	82.9	33.4	80.8	32.0	78.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	13.2	11.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	25.0	26.3	34.8	50.6	62.4	66.3	69.9	69.5	60.6	47.2	34.9	27.7
	DBStd	18.80	10.15	11.13	9.82	7.23	7.09	5.60	5.50	6.10	7.76	8.45	10.18	8.79
	HDD50	3308	776	663	475	82	3	0	0	0	8	155	455	691
	HDD65	6709	1241	1083	937	432	139	49	18	24	175	552	903	1156
	CDD50	2599	0	0	3	100	388	490	615	606	327	68	2	0
	CDD65	525	0	0	0	1	59	89	168	164	44	0	0	0
	CDH74	4719	0	0	0	15	493	682	1494	1620	412	3	0	0
	CDH80	1438	0	0	0	0	79	168	466	611	114	0	0	0
Wind	WSAvg	5.7	6.3	6.1	6.7	6.3	5.3	5.0	5.1	4.7	5.3	5.4	5.9	6.4
Precipitation	PrecAvg	20.5	1.8	1.5	1.3	1.5	1.8	2.2	2.1	1.8	1.4	1.6	1.7	2.0
	PrecMax	33.0	5.0	3.2	3.2	3.5	4.7	5.7	4.8	6.6	3.0	7.3	5.0	5.3
	PrecMin	10.9	0.3	0.1	0.1	0.1	0.5	0.2	0.3	0.1	0.1	0.0	0.2	0.2
	PrecStd	5.0	1.1	0.8	0.7	0.9	1.0	1.2	1.2	1.3	0.9	1.7	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	50.0	63.0	75.8	85.0	89.0	91.5	94.4	87.7	75.0	59.5	48.9
		MCWB	40.9	46.4	49.5	57.7	62.8	69.0	71.0	67.2	68.1	59.2	54.1	46.5
	2%	DB	39.6	44.6	56.5	70.5	82.0	84.4	88.1	90.3	83.1	69.0	53.9	43.1
		MCWB	37.6	42.0	47.0	55.1	62.0	67.4	68.8	67.6	64.2	57.1	50.5	41.0
	5%	DB	37.3	40.8	51.5	67.0	79.6	80.8	84.9	86.4	78.4	64.2	50.3	40.8
		MCWB	35.6	38.2	44.8	53.4	62.2	66.2	67.3	66.3	61.8	55.7	47.2	39.2
	10%	DB	35.7	38.0	47.1	63.3	76.3	77.6	81.6	82.7	74.2	59.7	46.9	37.3
		MCWB	34.4	35.8	41.8	51.2	61.9	64.7	66.1	65.4	60.2	54.2	44.4	35.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	46.7	51.1	60.0	67.9	72.2	74.0	72.4	69.4	62.7	55.1	46.7
		MCDB	42.1	49.3	60.6	73.1	78.1	82.5	87.5	85.7	85.6	69.5	58.9	48.4
	2%	WB	37.9	42.6	47.9	56.5	65.7	69.7	71.5	69.6	65.7	59.7	50.8	41.4
		MCDB	39.1	44.3	55.4	67.3	76.5	81.1	84.6	83.9	78.5	64.8	53.4	42.8
	5%	WB	35.9	38.6	45.2	54.6	64.2	67.8	69.3	67.8	63.3	57.2	47.7	38.9
		MCDB	36.8	40.2	50.5	64.6	75.0	78.2	80.1	82.2	75.8	62.0	49.7	40.9
	10%	WB	34.5	35.9	42.2	52.6	62.6	66.0	67.4	66.2	61.1	54.4	44.5	36.0
		MCDB	35.5	37.6	46.5	61.6	72.8	75.1	77.8	79.3	71.7	59.7	46.9	37.0
Mean Daily Temperature Range	5% DB	MDBR	7.9	10.0	12.5	16.3	18.4	17.2	18.7	19.8	18.1	14.8	9.8	7.6
		MCDBR	8.0	10.9	20.0	21.2	22.2	21.1	22.9	24.8	24.3	21.0	12.3	9.7
		MCWBR	7.4	8.9	12.2	10.7	9.0	9.5	8.5	9.2	10.2	12.1	9.1	8.6
	5% WB	MCDBR	7.2	10.0	18.1	18.9	19.2	18.2	19.4	21.6	22.7	16.8	10.5	9.2
	MCWBR	7.1	8.6	12.0	10.8	9.4	9.3	9.2	10.0	11.0	11.9	8.6	8.3	
Clear-Sky Solar Irradiance	taub	0.323	0.353	0.384	0.448	0.414	0.406	0.437	0.432	0.405	0.367	0.351	0.326	
	taud	2.195	2.151	2.196	2.084	2.267	2.310	2.228	2.228	2.272	2.353	2.301	2.271	
	Ebn at Noon	222	244	259	253	269	272	260	255	249	239	212	205	
	Edh at Noon	25	33	38	47	41	39	42	40	35	27	23	21	
All-Sky Solar Radiation	RadAvg	297	566	946	1406	1838	1970	1931	1726	1180	692	295	201	
	RadStd	57	101	135	129	155	170	118	116	154	103	53	44	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+1.37	N/A	N/A	+1.87	N/A	N/A	-265	-360	+220	+139	

Nomenclature: See separate page